

Last login: Thu Jun 18 19:58:22 on ttys003
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# SET ROLE alice;

SET

tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT COUNT(*)

FROM (

SELECT l_orderkey, l_linenum
FROM lineitem_secure
WHERE l_orderkey > 2000000

) l_alias;

WARNING: Log 1: Planner Input:

DETAIL: {QUERY

:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (

{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenum")
}

:rtekind 1

:subquery

{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (

{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum"
"l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnfl
ag" "l_linestatus" "l_shipdate" "l_commitdate" "l_receiptdate"
"l_shipinstruct" "l_shipmode" "l_comment")
}

:rtekind 1

:subquery

{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (

{RANGETBLENTY
:alias
{ALIAS
:aliasname l
:colnames <>
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate"
"l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}

:rtekind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}

:rteperminfos (

{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
:insertedCols (b)
:updatedCols (b)
}

:jointree

{FROMEXPR
:fromlist (

{RANGETBLREF
:rtindex 1
}

:quals

{SUBLINK
:sublinkType 0
:subLinkId 0
:testexpr <>

```

:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_supkey" "s_name" "s_address" "s_n
ationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh true
:relkind R
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{OEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:varatype 701
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcresultset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -24 3 0 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>

```

s.s_acctbal > 1000

```

        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDeps <>
        :withCheckOptions <>
        :stmt_location -1
        :stmt_len -1
    }
    :location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname l_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname l_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname l_supplekey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname l_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
}

```

```

{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13

```

```

:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 17016
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCI true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 17016
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointrac
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 130
}
{CONST
:consttype 23

```

L_orderkey > 2000000

```

        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 143
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
    :location 141
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location 70
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 17016
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location 82
    }
    :resno 2
    :resname l_linenummer
    :ressortgroupref 0
    :resorigtbl 17016
    :resorigcol 4
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos <>
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 0
      :aggargtypes <>
      :aggdirectargs <>
      :args <>
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 0
      :aggno -1
      :aggtransno -1
      :location 43
    }
    :resno 1
    :resname count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>

```

```

:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: max_parallel_hazard_walker: FuncExpr funcid=316 propparallel=s (s=safe r=restricted u=unsafe)
 WARNING: Log 2 : After pull up At starting of Path Generation:

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTTRY
  :alias
  {ALIAS
  :aliasname s
  :colnames <>
  }
  :eref
  {ALIAS
  :aliasname s
  :colnames ("s_supkey" "s_name" "s_address" "s_nationkey" "s_phone" "
s_acctbal" "s_comment")
  }
  :rtekind 0
  :relid 16541
  :inh false
  :relkind r
  :rellockmode 1
  :perminfoindex 1
  :tablesample <>
  :lateral false
  :inFromCl true
  :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
  :relid 16541
  :inh true
  :requiredPerms 2
  :checkAsUser 10
  :selectedCols (b 13)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals (
  {OPEXPR
  :opno 674
  :opfuncid 297
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 6
    :vartype 701
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location -1
    }
    {CONST
    :consttype 701
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
    }
  )
  :location -1
  }
)
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OPEXPR :opno 674 :opfuncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval true :constisnull false :location -1 :constvalue 8 [0 0 0 0 0 64 -113 64]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_s_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 1) :required_relds (b 1) :incompatible_relds (b) :outer_relds (b) :left_relds (b 1) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})

```

WARNING: is_parallel_safe: OUTPUT qual = ({RESTRICINFO :clause {OEXPR :opno 674 :opfncid 297 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 1 :varattno 6 :vartype 701 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 1 :varattnosyn 6 :location -1} {CONST :consttype 701 :consttypmod -1 :constcollid 0 :constlen 8 :constbyval t :constisnull false :location -1 :constvalue 8 [ 0 0 0 0 64 -113 64 ]}) :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relids (b 1) :required_relids (b 1) :incompatible_relids (b) :outer_relids (b) :left_relids (b 1) :right_relids (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeofamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL: (PATH
:pathtype 339
:parent_relds (b 1)
:pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs ()
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
})
:required_outer (b)
:parallel_aware false
:parallel_safe true
:parallel_workers 0
:rows 8196
:disabled_nodes 0
:startup_cost 0
:total_cost 343
:pathkeys <>
})

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: (QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17016
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname l
:colnames <>
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:teperminfos (
{RTEPERMISSIONINFO
:relid 17016
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
)

```

```

)
:jointree
{FROMEXPR
:fromlist (
:FROMEXPR
:fromlist (
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 3
}
)
):quals (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
}
):quals (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcoid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
){CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 143
:constvalue 4 [-128 -124 30 0 0 0 0 0]
}
):location 141
}
)
):quals <>
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)

```

I_orderkey > 2000000

```

WARNING: is_parallel_safe: INPUT qual = ({RESTRUCTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocons
tant true :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relds (b) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left
_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :h
ashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opres
ulttype 16 :opretset false :opcoid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 1 :varatype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattno
syn 1 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 0]}) :location -1} :is_pushed_d
own true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompati
ble_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_e
m <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: max_parallel_hazard_walker: Param paramid=0 in_safe_list=1 → reason=outer_reference
WARNING: is_parallel_safe: OUTPUT qual = ({RESTRUCTINFO :clause {PARAM :paramkind 1 :paramid 0 :paramtype 16 :paramtypmod -1 :paramcollid 0 :location -1} :is_pushed_down true :can_join false :pseudocon
stant true :has_clone false :is_clone false :has_volatile 0 :security_level 0 :num_base_rels 0 :clause_relds (b) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left
_relds (b) :right_relds (b) :orclause <> :rinfo_serial 1 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :h
ashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0} {RESTRUCTINFO :clause {OPEXPR :opno 521 :opfuncid 147 :opre
sulttype 16 :opretset false :opcoid 0 :inputcollid 0 :args ({VAR :varno 3 :varattno 1 :varatype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattno
syn 1 :location -1} {CONST :consttype 23 :consttypmod -1 :constcollid 0 :constlen 4 :constbyval true :constisnull false :location -1 :constvalue 4 [-128 -124 30 0 0 0 0 0]}) :location -1} :is_pushed_
down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompat
ible_relds (b) :outer_relds (b) :left_relds (b 3) :right_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_e
m <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe =
true | Reason = PROP_PARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROP_PARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilt
er <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2803 :aggtype 20 :aggcollid 0 :inputcollid 0 :aggtranstype 20 :aggargtypes <> :aggdirectargs <> :args <> :aggorder <> :aggdistinct <> :aggfilt

```

```

ter <> :aggstar true :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggssplit 0 :aggno 0 :aggtransno 0 :location -1)) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: Log 3: At the starting of Plan creation:
DETAIL:
{AGGPATH
:path.pathtype 365
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 95477.20104887121
:path.total_cost 95477.2110488712
:path.pathkeys <>
:subpath
{GATHERPATH
:path.pathtype 368
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 2
:path.disabled_nodes 0
:path.startup_cost 95476.94419918844
:path.total_cost 95477.15419918843
:path.pathkeys <>
:subpath
{AGGPATH
:path.pathtype 365
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 1
:path.disabled_nodes 0
:path.startup_cost 94476.94419918844
:path.total_cost 94476.95419918843
:path.pathkeys <>
:subpath
{INDEXPATH
:path.pathtype 342
:parent_relids (b 3)
:path.pathtarget
{PATHTARGET
:exprs <>
:sortgrouprefs <>
:cost.startup 0
:cost.per_tuple 0
:width 0
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware true
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 1661347
:path.disabled_nodes 0
:path.startup_cost 0.4325
:path.total_cost 90323.57669918844
:path.pathkeys <>
:indexinfo
{INDEXOPTINFO
:indexoid 17007
:reltablespace 0
:pages 16457
:tuples 6001215
:tree_height 2
:ncolumns 2
:nkeycolumns 2
:indexkeys ( 1 4)
:indexcollations ( 0 0)
:optfamily ( 1976 1976)
:opcintype ( 23 23)
:sortopfamily ( 1976 1976)
:reverse_sort ( false false)
:nulls_first ( false false)
:canreturn ( true true)
:relam 403
:indpred <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:indrestrictinfo (
{RESTRICTINFO
:clause
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
}
:is_pushed_down true
:can_join false

```

```

:pseudoconstant true
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 0
:clause_relds (b)
:required_relds (b 3)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b)
:right_relds (b)
:orclause <>
:rinfo_serial 1
:eval_cost.startup 0
:eval_cost.per_tuple 0
:norm_selec -1
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
)
{RESTRICTINFO
:clause
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 143
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 141
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relds (b 3)
:required_relds (b 3)
:incompatible_relds (b)
:outer_relds (b)
:left_relds (b 3)
:right_relds (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
)
}
:predOK false
:unique false
:nullsnotdistinct false
:immediate true
:hypothetical false
:amcanorderbyop false
:amoptionalkey true
:amsearcharray true
:amsearchnulls true
:amhasgettupel true
:amhasgetbitmap true
:amcanparallel true
:amcanmarkpos true
}
:indexclauses (
{INDEXCLAUSE
:rinfo
{RESTRICTINFO
:clause
{OEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false

```

```

        :location 143
        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
}
:location 141
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relids (b 3)
:required_relids (b 3)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 3)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
:indexquals (
{RESTRICINFO
:clause
{OPEXPR
:opno 521
:opfncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 143
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
}
:location 141
}
:is_pushed_down true
:can_join false
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 0
:security_level 0
:num_base_rels 1
:clause_relids (b 3)
:required_relids (b 3)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 3)
:right_relids (b)
:orclause <>
:rinfo_serial 2
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 0.6644041354651555
:outer_selec -1
:mergeopfamilies <>
:left_em <>
:right_em <>
:outer_is_left false
:hashjoinoperator 0
:left_bucketsize -1
:right_bucketsize -1
:left_mcvfreq -1
:right_mcvfreq -1
:left_hasheqoperator 0
:right_hasheqoperator 0
}
}
:lossy false
:indexcol 0
:indexcols <>
}
)
:indexorderby <>
:indexorderbycols <>
:indexscandir 1
:indextotalcost 73644.6725
:indexselectivity 0.6644041354651555
}
:aggstrategy 0
:aggssplit 6
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}
:single_copy false
:num_workers 2
}
:aggstrategy 0
:aggssplit 9
:numGroups 1
:transitionSpace 0
:groupClause <>
:qual <>
}
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:
DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false

```

:dependsOnRole false
:parallelModeNeeded true
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 95477.20104887121
:plan.total_cost 95477.2110488712
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 9
:aggno 0
:aggtransno 0
:location 43
}
:resno 1
:resname count
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{GATHER
:plan.disabled_nodes 0
:plan.startup_cost 95476.94419918844
:plan.total_cost 95477.15419918843
:plan.plan_rows 2
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 1
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 20
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 94476.94419918844
:plan.total_cost 94476.95419918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location 43
}
:resno 1
:resname <>
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
}

```

```

:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{RESULT
:plan.disabled_nodes 0
:plan.startup_cost 0.4325
:plan.total_cost 98323.57669918844
:plan.plan_rows 1661347
:plan.plan_width 0
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist <>
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 98323.57669918844
:scan.plan.plan_rows 1661347
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual <>
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 1)
:scan.plan.allParam (b 1)
:scan.scanrelid 3
:indexid 17007
:indexqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 143
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location 141
)
)
:recheckqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 143

```

```

        :constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
    }
    }
    :location 141
}
)
:indexorderby <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:resconstantqual (
{PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0
:location -1
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggssplit 6
:numcols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:num_workers 2
:rescan_param 1
:single_copy false
:invisible false
:initParam (b 0)
}
:plan.righttree <>
:plan.initPlan (
{SUBPLAN
:sublinkType 0
:testexpr <>
:paramIds <>
:plan_id 1
:plan_name InitPlan\ 1
:firstColType 2278
:firstColTypeMod -1
:firstColCollation 0
:useHashTable false
:unknownEqFalse true
:parallel_safe true
:setParam (i 0)
:parParam <>
:args <>
:startup_cost 0
:per_call_cost 0.041849682772083945
}
)
:plan.extParam (b)
:plan.allParam (b 0)
:aggstrategy 0
:aggssplit 9
:numcols 0
:grpColIdx ()
:grpOperators ()
:grpCollations ()
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumbers")
}
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0

```

```

:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17016
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname 1
:colnames <>
}
:eref
{ALIAS
:aliasname 1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 2 3 4)
:permInfos (
{RTEPERMISSIONINFO
:relid 17016
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans (
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 343
:scan.plan.plan_rows 8196
:scan.plan.plan_width 0
:scan.plan.parallel_aware false
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 5
:scan.plan.targetlist <>
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 6
:varatype 701
:varitypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1

```

```

:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b)
:scan.plan.allParam (b)
:scan.scanrelid 4
}
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 17016 16566 16541)
:invalidItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: (PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 94476.94419918844
:plan.total_cost 94476.95419918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{RESULT
:plan.disabled_nodes 0
:plan.startup_cost 0.4325
:plan.total_cost 98323.57669918844
:plan.plan_rows 1661347
:plan.plan_width 0
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist <>
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 98323.57669918844
:scan.plan.plan_rows 1661347
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:vartype 23
:varitypmode -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
}
}
}
}

```

```

:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual <>
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 1)
:scan.plan.allParam (b 1)
:scan.scanrelid 3
:indexid 17007
:indexqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
:recheckqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
:indexorderby <>
:indexlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 4
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:indexorderdir 1
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:resconstantqual (
(PARAM
:paramkind 1
:paramid 0
:paramtype 16
:paramtypmod -1
:paramcollid 0

```

```

        :location -1
    }
}
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0 1)
:plan.allParam (b 0 1)
:aggstrategy 0
:aggsplit 6
:numCols 0
:grpColIdx <>
:grpOperators <>
:grpCollations <>
:numGroups 1
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable {
{RANGETBENTRY
:alias
{ALIAS
:aliasname l_alias
:colnames <>
}
:eref
{ALIAS
:aliasname l_alias
:colnames ("l_orderkey" "l_linenumber")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 0
:inh false
:relkind <>
:rellockmode 0
:perminfoindex 0
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_secure
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 17016
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname l
:colnames <>
}
:eref
{ALIAS
:aliasname l
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname s
:colnames <>
}
:eref
{ALIAS
:aliasname s
:colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
}
:rtekind 0
:relid 16541
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 2 3 4)
:permInfos {
{RTEPERMISSIONINFO
:relid 17016
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 11)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16541
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 13)
:insertedCols (b)
:updatedCols (b)
}
}
:resultRelations <>
:appendRelations <>

```

```

:subplans (
  {SEQSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 343
    :scan.plan.plan_rows 8196
    :scan.plan.plan_width 0
    :scan.plan.parallel_aware false
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 5
    :scan.plan.targetlist <>
    :scan.plan.qual (
      {OPEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 4
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 6
            :location -1
          }
          {CONST
            :consttype 701
            :consttypmod -1
            :constcollid 0
            :constlen 8
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
          }
        )
        :location -1
      }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b)
    :scan.plan.allParam (b)
    :scan.scanrelid 4
  }
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL:  {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 0
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 94476.94419918844
:plan.total_cost 94476.95419918843
:plan.plan_rows 1
:plan.plan_width 8
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 6
:aggno 0
:aggtransno 0
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{RESULT
:plan.disabled_nodes 0
:plan.startup_cost 0.4325
:plan.total_cost 90323.57669918844
:plan.plan_rows 1661347
:plan.plan_width 0
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist <>
:plan.qual <>
:plan.lefttree
{INDEXONLYSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0.4325
:scan.plan.total_cost 90323.57669918844
:scan.plan.plan_rows 1661347
:scan.plan.plan_width 0
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 4
:scan.plan.targetlist (

```

```

{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -3
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 4
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual <>
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extraParam (b 1)
:scan.plan.allParam (b 1)
:scan.scanrelid 3
:indexid 17007
:indexqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
)
:recheckqual (
{OPEXPR
:opno 521
:opfuncid 147
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -128 -124 30 0 0 0 0 0 ]
}
)
:location -1
}
)
)
:indexorderby <>
:indextlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}

```

```

    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 3
    :varattno 4
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 4
    :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    }
    :indexorderdir 1
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0 1)
    :plan.allParam (b 0 1)
    :resconstantqual (
    {PARAM
    :paramkind 1
    :paramid 0
    :paramtype 16
    :paramtypmod -1
    :paramcollid 0
    :location -1
    }
    )
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0 1)
    :plan.allParam (b 0 1)
    :aggstrategy 0
    :aggsplit 6
    :numCols 0
    :grpCollIdx <>
    :grpOperators <>
    :grpCollations <>
    :numGroups 1
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :partPruneInfos <>
    :rtable (
    {RANGETBENTRY
    :alias
    {ALIAS
    :aliasname l_alias
    :colnames <>
    }
    }
    :eref
    {ALIAS
    :aliasname l_alias
    :colnames ("l_orderkey" "l_linenumber")
    }
    }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 0
    :inh false
    :relkind <>
    :rellockmode 0
    :perminfoindex 0
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBENTRY
    :alias <>
    :eref
    {ALIAS
    :aliasname lineitem_secure
    :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
    }
    }
    :rtekind 1
    :subquery <>
    :security_barrier false
    :relid 17016
    :inh false
    :relkind v
    :rellockmode 1
    :perminfoindex 1
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBENTRY
    :alias
    {ALIAS
    :aliasname l
    :colnames <>
    }
    }
    :eref
    {ALIAS
    :aliasname l
    :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
    }
    }
    :rtekind 0
    :relid 16566
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 2
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
    }
    {RANGETBENTRY
    :alias
    {ALIAS
    :aliasname s
    :colnames <>
    }
    }
    :eref
    {ALIAS
    :aliasname s
    :colnames ("s_suppkey" "s_name" "s_address" "s_nationkey" "s_phone" "s_acctbal" "s_comment")
    }
    }
    :rtekind 0
    :relid 16541
    :inh false
    :relkind r
    :rellockmode 1
    :perminfoindex 3
    :tablesample <>

```

```

:lateral false
:inFromCl true
:securityQuals <>
}
)
:unprunableRelids (b 2 3 4)
:permInfos (
  {RTEPERMISSIONINFO
    :relid 17016
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 11)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16541
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:resultRelations <>
:appendRelations <>
:subplans (
  {SEQSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 343
    :scan.plan.plan_rows 8196
    :scan.plan.plan_width 0
    :scan.plan.parallel_aware false
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 5
    :scan.plan.targetlist <>
    :scan.plan.qual (
      {OPEXPR
        :opno 674
        :opfuncid 297
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 4
            :varattno 6
            :vartype 701
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 6
            :location -1
          }
          {CONST
            :consttype 701
            :consttypmod -1
            :constcollid 0
            :constlen 8
            :constbyval true
            :constisnull false
            :location -1
            :constvalue 8 [ 0 0 0 0 0 64 -113 64 ]
          }
        )
        :location -1
      }
    )
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b)
    :scan.plan.allParam (b)
    :scan.scanrelid 4
  }
)
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 16 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

QUERY PLAN

```

Finalize Aggregate  (cost=95477.20..95477.21 rows=1 width=8) (actual time=329.877..331.425 rows=1.00 loops=1)
  Output: count(*)
  Buffers: shared hit=10991
  InitPlan 1
    -> Seq Scan on public.supplier s  (cost=0.00..343.00 rows=8196 width=0) (actual time=0.032..0.032 rows=1.00 loops=1)
      Filter: (s.s_acctbal > '1000'::double precision)
      Buffers: shared hit=2
    -> Gather  (cost=95476.94..95477.15 rows=2 width=8) (actual time=329.778..331.419 rows=3.00 loops=1)
      Output: (PARTIAL count(*))
      Workers Planned: 2
      Workers Launched: 2
      Buffers: shared hit=10991
      -> Partial Aggregate  (cost=94476.94..94476.95 rows=1 width=8) (actual time=324.126..324.126 rows=1.00 loops=3)
        Output: PARTIAL count(*)
        Buffers: shared hit=10989
        Worker 0:  actual time=320.627..320.628 rows=1.00 loops=1
          Buffers: shared hit=3633
        Worker 1:  actual time=322.376..322.376 rows=1.00 loops=1
          Buffers: shared hit=3664
      -> Result  (cost=0.43..90323.58 rows=1661347 width=0) (actual time=0.174..0.267.340 rows=1333572.67 loops=3)
        One-Time Filter: (InitPlan 1).coll1
        Buffers: shared hit=10989
        Worker 0:  actual time=0.080..0.264.336 rows=1324101.00 loops=1
          Buffers: shared hit=3633
        Worker 1:  actual time=0.071..0.265.558 rows=1335534.00 loops=1
          Buffers: shared hit=3664
      -> Parallel Index Only Scan using 1_pk on public.lineitem l  (cost=0.43..90323.58 rows=1661347 width=0) (actual time=0.172..0.168.399 rows=1333572.67 loops=3)
        Output: l.l_orderkey, l.l_linenumber
        Index Cond: (l.l_orderkey > 2000000)
        Heap Fetches: 1
        Index Searches: 1
        Buffers: shared hit=10989
        Worker 0:  actual time=0.079..0.166.805 rows=1324101.00 loops=1
          Buffers: shared hit=3633
        Worker 1:  actual time=0.070..0.167.196 rows=1335534.00 loops=1
          Buffers: shared hit=3664
  Planning:
    Buffers: shared hit=107
  Planning Time: 13.889 ms
  Execution Time: 336.296 ms
(40 rows)

tpch=>

```